

Institute/School/College Name	Chitkara University Institute of Engineering & Technology		
Department/Centre Name	Department of Applied Sciences		
Programme Name	B.E.-CSE		
Course Name	Calculus and Statistical Analysis	Session	2023 - 2024
Course Code	22AS001	Semester/Batch	I st /2023
Lecture/Tutorial (Per Week)	4-0-0	Course Credit	4
Course Coordinator Name	Dr. Madhu Aneja		

1. Objective of the Course:

- To provide the ability to apply mathematics for the solution of complex engineering and real life problems.
- To Identify, formulate and analyse the engineering problems.
- To arrive at substantiated conclusions using principles of mathematics are covered.

The main objectives of the courses are:

- To design matrices for mathematical problems related to real life and find their solutions.
- To analyse functions of two or more variables and compute their derivatives for finding extreme values of surfaces.
- To apply techniques of evaluating double and triple integral to solve various engineering problems.
- To implement various hypothesis testing techniques for small and large sample data and calculate coefficient of correlation, line of regression to describe relation between independent variable and dependent variable.

2. Course Learning Outcome:

	Course Outcome	POs	CL	KC	Sessions
CLO01	Students will be able to introduce and form matrices to present mathematical solutions in a concise and informative manner. Use matrices to solve the problems of system of linear equations and solve various live problems using matrices.	PO1,PO3,PO10	K2	Factual Conceptual	12
CLO02	Students will be able to find local extreme values of functions of several variables, test for saddle points, examine the conditions for the existence of absolute extreme values. Solve constraint problems using Lagrange multipliers and solve related application problems.	PO1,PO2,PO3,PO4,PO12	K3	Fundamental Conceptual	15
CLO03	Students will be able to apply the principles of Integral Calculus to solve a variety of practical	PO3,PO5,PO12	K3	Conceptual Procedural	9

	problems in Engineering and applied Sciences.				
CLO04	Students will be able to interpret statistical inference tasks with the help of probability & distributions and hypothesis testing for means, variances and proportions of large as well as small data and employ appropriate regression models in determining statistical relationships.	PO2,PO3,PO5,PO9,PO11,PO12	K3	Conceptual Procedural	9
Total Contact Hours					45

Revised Bloom's Taxonomy Terminology

*Cognitive Level =CL

*Knowledge Categories = KC

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1	H	H		H								
CLO2	H	H										
CLO3	H		M									H
CLO4	H	H										M

3. ERISE Grid Mapping

Feature Enablement	Level(1-5, 5 being highest)
Entrepreneurship	2
Research	4
Innovation	3
Skills	5
Employability	4

4. Recommended Books (Reference Books/Text Books):

- B01:** Advanced Engineering Mathematics, Erwin Kreyszig, Wiley India Pvt. Ltd
B02: Engineering Mathematics, Srimanta Pal &Subodh C. Bhunia, Edition 2015, Oxford University Press.
B03: The Engineering Mathematics, 2nd Edition, Chitkara University Publication, Vol. I.
B04: Higher Engineering Mathematics, B.V. Ramana, Tata McGraw-Hill Education.
B05: Advanced Engineering Mathematics, R.K. Jain and S.R.K. Iyengar, Alpha Science International Ltd.
B06: Higher Engineering Mathematics, B.S. Grewal, Khanna Publications.
B07: A text book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications.

B08: Calculus, by Howard Anton, Irl Bivens Stephens Davis

B09: Advanced Engineering Mathematics, H.C. Taneja, I.K. International, Vol I.

B10: Introduction to Probability and Statistics for Engineers and Scientists 4th edition, Sheldon M. Ross, Academic Press, Elsevier.

5. Other readings & relevant websites:

S.N.	Link of Journals, Magazines, websites and Research Papers
1.	http://www.nptelvideos.com/video.php?id=67&c=1
2.	http://www.nptelvideos.com/video.php?id=71&c=1
3.	http://www.nptelvideos.com/video.php?id=91&c=1
4.	http://mathinsight.org/partial_derivative_examples
5.	http://www.studyaaar.com/index.php/module/58-multiple-integrals-and-their-applications
6.	www.intmath.com/Matrices-determinants/Matrix-determinant-intro.php
7.	http://nptel.ac.in/courses/111105041/23

6. Course Plan:

Session No.	Topics	Recommended Books
1	Prerequisite Differentiation: Geometrical interpretation of derivative, Indefinite and definite (integration by substitution, by parts, by partial fraction) Reduction formulae sine and cosine (with limit $0 - \pi/2$), Review of matrices and determinants, Elementary operations	B6
1 -3	Matrices: Rank, Inverse of matrix, Normal form	B6
3 - 4	Solution of a system of linear equations by using rank,	B6
4- 5	Characteristics equations , Eigen values and Eigen vectors	B7
5 - 7	Cayley Hamilton theorem(without proof), Diagonalization,	B7
7- 8	Quadratic form & Canonical form.	B7
8 - 9	Partial Derivatives: Partial Differentiation, Euler's Theorem for homogeneous functions.	B6
9 - 10	Tangent and Normal, Partial Derivative of Composite Functions, Implicit Functions	B6
10 - 12	Total derivatives, Error & Approximation.	B6
ST-1		
12 - 13	Jacobians with properties, Taylor's Series Expansion, Maclaurin's Series (one & two variables).	B4
13- 15	Maxima and Minima of functions of two and three variables	B4
15 - 16	Lagrange's method of Undetermined Multipliers	B4
16- 18	Sketch some standard Cartesian and Polar curves using Curve tracing - Cardioid, Lemniscate, Folium of Descartes, Three/Four Leaved Rose, Limacon, Cissoid	B4
18 - 19	Multiple Integration : Reduction formulas, Double Integration using Cartesian & polar coordinate	B4
19- 20	Change of order in double integration, Triple Integration	B4
20- 21	Change of variables in Polar, Cylindrical and Spherical Coordinates	B4
21 - 22	Area enclosed by Plane curves	B4

22 -23	Volume, Moment of Inertia, Centroid and Centre of Gravity	B4
23 – 24	Improper integrals of first and second kind, Beta and Gamma functions	B4
ST-2		
24 – 25	Probability and Statistical Methods: Random variable(discrete and continuous), Binomial, Poisson	B5
25 – 26	Normal, exponential distribution.	B5
26- 27.5	Hypothesis Testing (General concepts, Testing a Statistical hypothesis, one and two tailed tests, Critical region, Confidence interval estimation, Single and two sample tests on proportion), mean and variance	B5
27.5 – 29	Sampling Distribution of means and variance, t-distribution and F-distribution,	B5
29 - 30	Correlation ,lines of regression(two variable only)	B5
END TERM – FULL SYLLABUS		

7. Delivery/Instructional Resources

Lecture No.	Topics	PPT (link of ppts on the central server)	Industry Expert Session(If yes: link of ppts on the central server)	Web References	Audio-Video
1	Prerequisite: Differentiation: Geometrical interpretation of derivative, Indefinite and definite (integration by substitution, by parts, by partial fraction), Reduction formulae sine and cosine (with limit $0 - \pi/2$)	https://docs.google.com/presentation/d/1vrZY0sHDOg7ylrsnt7PZzQSWQNzbDS	NA	https://nptel.ac.in/courses/122/104/122104018/	https://www.youtube.com/watch?v=EJ8Ptnk7dzU&t=1002s
1-8	Matrices: Rank, Inverse of matrix, Normal form Solution of a system of linear equations by using rank Characteristics equations, Eigen values and vectors, Cayley Hamilton theorem(without proof) Diagonalization, Quadratic form & Canonical form,	https://docs.google.com/presentation/d/1vrZY0sHDOg7ylrsnt7PZzQSWQNzbDS	NA	https://nptel.ac.in/courses/122/104/122104018/	https://nptel.ac.in/courses/122/104/122104018/
8 - 16	Partial Differentiation, Euler's Theorem for homogeneous functions, Tangent and Normal, Partial Derivative of Composite Functions, Implicit Functions Total derivatives, Error & Approximation. Jacobians with properties, Taylor's Series Expansion, Maclaurin's Series (one & two variables).	https://docs.google.com/presentation/d/1vrZY0sHDOg7ylrsnt7PZzQSWQNzbDS	NA	https://nptel.ac.in/courses/111/107/111107108/	https://nptel.ac.in/courses/111/107/111107108/



	Maxima and Minima of functions of two and three variables Lagrange's method of Undetermined Multipliers				
16 – 18	Sketch some standard Cartesian and Polar curves using Curve tracing - Cardioid, Lemniscate, Folium of Descartes, Three/Four Leaved Rose, Limacon, Cissoid.	https://docs.google.com/presentation/d/1vrZY0sHDOg7ylrsnt7PZzQSWQNZbDS DU/edit	NA	https://www.youtube.com/watch?v=OAEjSiFGdhQ	https://www.youtube.com/watch?v=IAb98ZgSJNw
18 – 24	Multiple Integration: Reduction formulas, Double Integration using Cartesian & polar coordinate, Change of order in double integration, Triple Integration. Change of variables in Polar, Cylindrical and Spherical Coordinates. Area enclosed by Plane curves, Volume, Moment of Inertia, Centroid and Centre of Gravity. Improper integrals of first and second kind, Beta and Gamma functions	https://docs.google.com/presentation/d/1vrZY0sHDOg7ylrsnt7PZzQSWQNZbDS DU/edit	NA	https://npTEL.ac.in/courses/111/107/111107108/	https://npTEL.ac.in/courses/111/107/111107108/
24 - 30	Probability and Statistical Methods: Random variable(discrete and continuous), Binomial, Poisson Normal, exponential distribution. Hypothesis Testing (General concepts, Testing a Statistical hypothesis, one and two tailed tests, Critical region, Confidence interval estimation, Single and two sample tests on proportion), mean and variance Sampling Distribution of means and variance, t-distribution and F-distribution, Correlation ,lines of regression(two variable only)	https://docs.google.com/presentation/d/1vrZY0sHDOg7ylrsnt7PZzQSWQNZbDS DU/edit	NA	https://npTEL.ac.in/courses/111/105/111105041/	https://npTEL.ac.in/courses/111/105/111105041/

8. Action plan for different types of learners

Slow Learners	Average Learners	Fast Learners
Extra Class on Saturday. Doubt sessions.	Doubt-sessions	Doubt Sessions

9. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Component 1	FA	01	10%	Offline
Component 2	Sessional Tests (STs)	02*	30%	Online/Offline
Component 3	End Term Examinations	01	60%	Online/Offline
Total		100%		

* Out of 2 ST's the ERP system automatically picks the best one ST's marks for internal assessment.

Details of Evaluation Components:

Evaluation Component	Description	Syllabus Covered (%)	Timeline of Examination	Weightage (%)
Component 01	FA	20%	Before ST1	10%
Component 02	ST 01	Up to 40%	As defined in Academic Calendar	30%
	ST 02	41% - 80%	As defined in Academic Calendar	
Component 03	End Term Examination*	100%	At the end of the semester	60%
Total				100%

*As per Academic Guidelines minimum 75% attendance is required to become eligible for appearing in the End Semester Examination.

Evaluation Components of Sessional Test and End Term Examination

Type of Assessment	Time of Conduction	Total Marks
Sessional Test 1	1-12 Sessions	40
Sessional Test 2	12 - 24 Sessions	40
End Term Examination		60

10. Syllabus of the Course:

Name of the course: Calculus and Statistical Analysis		Subject Code: 22AS001	
Contents		No. of Sessions	Weightage (%)
Unit 1 Matrices: Rank, Inverse of matrix, Normal form, Solution of a system of linear equations by using rank, Characteristics equations, Eigen values and vectors, Cayley Hamilton theorem(without proof), Diagonalization, Canonical form, Quadratic form.		8	17.5

Unit 2 Partial Differentiation: Partial Differentiation, Euler's Theorem for homogeneous functions, Total derivatives, Error & Approximation, Tangent and Normal. Partial Derivative of Composite Functions, Implicit Functions, Jacobians (with properties), Taylor's Series Expansion, Maclaurin's Series (one and two variables). Maxima and Minima of functions of two and three variables, Lagrange's method of Undetermined Multipliers.	8	27.5
Unit 3 CurveTracing: Curve tracing (Cartesian and polar curves)- Cissoid, cardioid, Lemniscate, Folium of Descartes, Three/Four Leaved Rose, Limacon, Multiple Integration: Double Integration using Cartesian & polar coordinate, Change of order in double integration, Triple Integration ,Change of variables in Polar, Cylindrical and Spherical Coordinates, Area enclosed by Plane curves, Volume, Moment of Inertia, Centroid and Center of Gravity, Improper integrals of first and second kind, Beta and Gamma functions	8	30
Unit 4 Probability and Statistical Methods: Random variable (discrete and continuous), Binomial, Poisson, Normal, exponential distribution. Sampling Distribution of means and variance, t-distribution and F- distribution, Hypothesis Testing (General concepts, Testing a Statistical hypothesis, one and two tailed tests, Critical region, Confidence interval estimation. Single and two sample tests on proportion), mean and variance .correlation ,lines of regression(two variable only)	6	25

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Madhu Aneja	
Program Incharge	Dr. Reetu Malhotra	
Dean	Dr. Mohit Kumar Kakkar	
DD/MM/YYYY	10/08/2023	